DATA 202 — Quiz 3

Week 6 — Covering Weeks 4–5: Joins, Tidy Data, Clustering & PCA

Name: _____ Date: _____

The Dataset: Campus Gym

The campus gym keeps three related tables (simulated data). Every question on this quiz uses them.

- `members` — one row per gym member: `member_id`, the `plan` they signed up for (typed by hand at the front desk, so the **raw** text is messy), and their habits per visit: `avg_minutes` (minutes per visit), `cardio_pct` (% of that time on cardio machines) and `calories` (calories burned per visit).
- `checkins` — how many times a member came to the gym in each week (`week_1`–`week_4`) of a `month`.
- `plans` — the gym's membership plans and their `monthly_fee` (dollars).

Question 7 works with the raw `plan` text. All other questions assume `plan` has already been cleaned to `"basic"`, `"premium"` or `"student-monthly"`.

First few rows of `members` (raw):

member_id	plan	avg_minutes	cardio_pct	calories
M001	Basic	44	70	390
M002	BASIC	51	71	420
M003	Prem	34	43	290
M004	basic	82	12	440
M005	Student - Monthly	45	98	440

First few rows of `checkins`:

member_id	month	week_1	week_2	week_3	week_4
M002	Sep	6	5	6	2
M002	Oct	6	1	2	4
M003	Sep	2	1	0	0
M003	Oct	0	0	0	1
M004	Sep	5	2	1	4
M006	Sep	4	3	3	4

All of `plans`:

plan	monthly_fee
basic	25
premium	45
student-monthly	15

Question 1 (SLO 04A)

For each column below, say whether it is a **primary key**, a **foreign key**, or **neither** in that table. For a foreign key, name the table it points to.

Table	Column	Primary key / Foreign key / Neither	If foreign key: points to which table?
members	member_id		
members	plan		
plans	plan		
checkins	member_id		
checkins	week_1		

Question 2 (SLO 04B)

Suppose `members` and `checkins` contained **only the rows shown on page 1**. How many rows would each join below return?

```
pd.merge(checkins, members, on="member_id", how=___)
```

how=	Number of rows
"inner"	
"left"	
"right"	
"outer"	

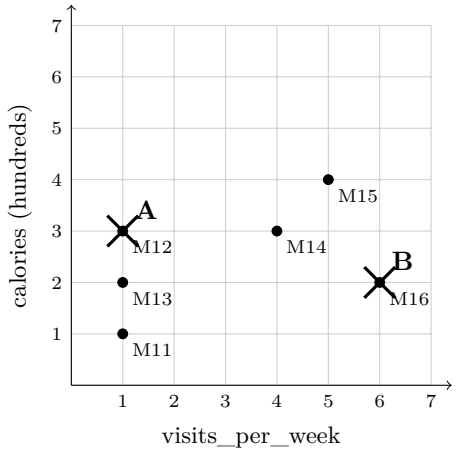
Question 3 (SLO 04C)

The code below reshapes `checkins` into a long table. Sketch the part of `checkins_long` that comes from **M002's two rows** of `checkins` (page 1): column names and every row.

```
checkins_long = pd.melt(
   checkins,
   id_vars=["member_id", "month"],
   value_vars=["week_1", "week_2", "week_3", "week_4"],
   var_name="week",
   value_name="visits",
)
```

Question 4 (SLO 05A)

Run **one round of k-means by hand** (k = 2) on these six members. The starting centroids are **A** = member M12 and **B** = member M16 (the crosses). First write the closest centroid for each member, then compute the two new centroids.



member	visits	calories (100s)	Closest centroid (A/B)
M11	1	1	
M12	1	3	
M13	1	2	
M14	4	3	
M15	5	4	
M16	6	2	

New centroid A = (_____ , _____) New centroid B = (_____ , _____)

Question 5 (SLO 05B)

For Questions 5–6, the gym built `profile`: one row per member (213 members) with `visits_per_week` (from `checkins`) plus `avg_minutes`, `cardio_pct` and `calories`, scaled into `X_scaled`. It ran k-means on `X_scaled` for k = 1 to 8 and recorded the inertia and the silhouette score of each run. **Which k would you choose?** Justify using both rows of the table.

k	1	2	3	4	5	6	7	8
inertia	852	436	165	139	116	96	88	82
silhouette	—	0.47	0.61	0.51	0.44	0.35	0.32	0.3

Question 6 (SLO 05C)

The gym ran PCA on the same four scaled columns (`X_scaled`) of `profile`:

```
PCA().fit(X_scaled).explained_variance_ratio_.round(3)
# [0.617 0.307 0.070 0.006]
```

How many components would you keep to retain at least **90%** of the variance? What share of the variance do they keep?

Question 7 (SLO 03A) — Retake from Quiz 2

The code below is run on the **raw** `plan` column of `members` (page 1). Write what each `plan` value becomes.

```
members["plan"] = (
   members["plan"]
   .str.strip()
   .str.lower()
   .str.replace(r"^prem$", "premium", regex=True)
)
```

member_id	plan (raw)	plan (after the code)
M001	Basic	
M002	BASIC	
M003	Prem	
M004	basic	
M005	Student - Monthly	

Question 8 (SLO 03B) — Retake from Quiz 2

Complete the code to compute the average `calories` for each `plan` (cleaned), sorted from highest to lowest.

```
result = (
   members.groupby("_____________")["_____________"]
   ._________()
   .reset_index()
   .sort_values("calories", ascending=_________)
)
```

Question 9 (SLO 03C) — Retake from Quiz 2

For each question below, name the best plot type (Histogram / Scatter / Line / Bar) and the column(s) you'd map to x (and y, if relevant).

Question	Plot Type	Column(s)
What's the distribution of calories across all members?		
How do avg_minutes and calories relate to each other?		
How does the total number of visits change from week_1 to week_4 (using checkins_long)?		
How does average cardio_pct compare across plan values?		